

Differential Gene Expression Analysis in Breast Cancer

Abstract

The objective of this project is to identify genes which are differentially expressed between tumor and normal breast tissue sample. RNA-seq data were acquired from the Cancer Genome Atlas (TCGA). 113 normal and tumor tissue samples were used to perform the differential expression analysis. R packages which were used for the analysis were TCGAbiolinks and DESeq2. The findings identified breast cancer markers such as TSPAN6, TNMD, and CFTR.

Introduction

Amongst women, Breast Cancer is one of the leading and deadly cancers. In order to prevent, or treat these conditions, it is important to understand the molecular differences between normal and cancerous tissues. Targeted therapy and early diagnostic tools can make use of the biomarkers found from this study. This project looks into the genes which are significantly different from the normal and tumor tissue samples. The TCGA-BRCA database allows us to visualize the difference through a large-scale transcriptomic comparison. The main objective is to identify the therapeutic targets and potential biomarkers through differential gene expression analysis.

Materials and Methods

Databases and Tools

1. **Dataset:** TCGA-BRCA RNA-Seq data (STAR - Counts)
2. **Programming Language:** R
3. **Key Packages:** TCGAbiolinks, DESeq2, SummarizedExperiment, EnhancedVolcano, biomaRt, pheatmap, ggplot2, dplyr

Methods

1. **Query and Download:** Used TCGAbiolinks to extract gene expression data for "Primary Tumor" and "Solid Tissue Normal" samples.
2. **Filtering:** Selected 113 patients with matched tumor-normal samples.
3. **Data Preparation:** Labeled samples by condition (Tumor vs Normal) and created DESeq2 dataset.
4. **Differential Expression:** Removed low-count genes, ran DESeq2, applied log-fold change shrinkage with apeglm.
5. **Visualization:** Generated PCA plots, top 20 gene heatmap (with gene symbol mapping using biomaRt), and gene clustering dendrogram.

Results

Through the analysis of the normal tissues and tumor tissues, three figures were generated: PCA, Heatmap and a Dendrogram. The principal component analysis (PCA) shows transcriptomic differences between the two tissue samples. There is a clear separation between the normal tissue and tumor tissue. Each dot is a tissue sample from a patient. Red dots represent normal tissue and blue represents tumor tissue. The axis combines many genes into one score to visually understand the patterns. PC1 which is principal component 1 explains 36% of the variation in the data set and PC2 shows 17% of variance. Both PC1 and PC2 axes show the biggest differences in data respectively. The split in the middle clearly shows tumor gene activity is significantly different from normal genes.

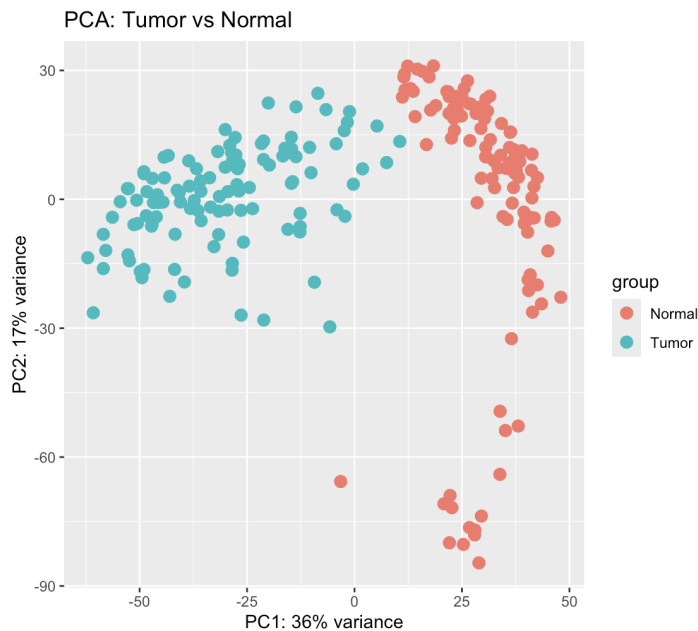


Figure 1: Principal Component Analysis: Tumor vs Normal

PCA plot visualizing gene expression profile from breast cancer sample. Red dots represent normal and blue dots represent tumor tissue samples. PC1 explains 36% of variance and PC2 represent 17% of variance. Clustering of blue dots on one side and red on the other highlight differences in gene activity between normal and cancerous tissue.

A heatmap was also produced to visualize the genes which were upregulated in a cancerous tissue compared to a normal tissue. Each column represents a tissue sample from a patient and each row represents a different gene. The condition bar represents normal tissue (blue-green) and tumor tissue (red). The gradient bar on the right from blue to red represents how much the gene is regulated. Red means the gene is upregulated and blue means the gene is downregulated. On the right 20 most differentially active genes were mentioned. The top 5 genes which illustrate key different gene expression are TSPAN6, TNMD, and CFTR. In TSPAN6 genes are way more upregulated in normal tissues compared to tumor tissues. In CFTR and TNMD, the tumor tissues are more downregulated compared to the normal tissue.

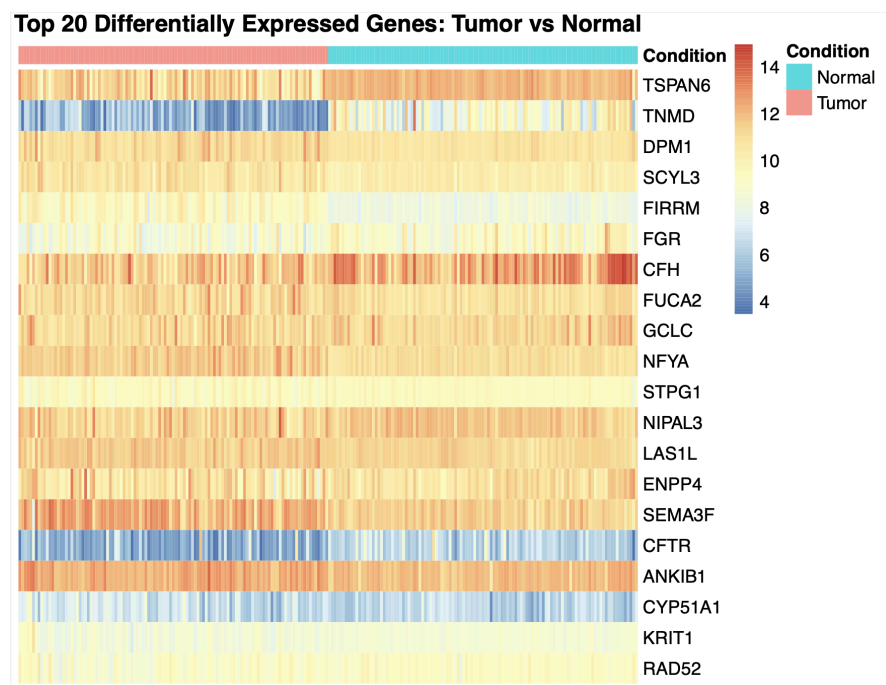


Figure 2: Heatmap of Top 20 Differentially Expressed Genes

Each row represents genes listed in the right side and each column represents a patient sample. The condition bar on top shows whether the patient sample is a tumor tissue (Blue-Green) or a normal tissue (Red). The gradient on the right from blue to red represents downregulation (Blue) or upregulation (Red).

A dendrogram is also generated which shows the similarities between the top 20 differentially expressed genes. TNMD, CFTR and CYP51A1 are closely clustered which means they have similar expression trends.

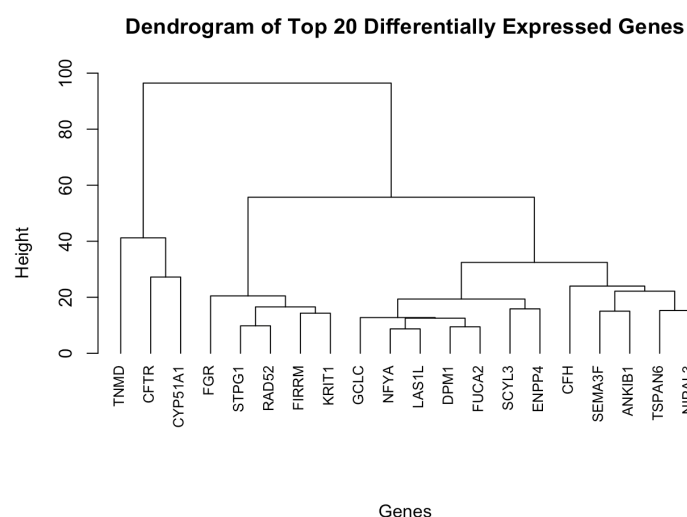


Figure 3: Dendrogram of Top 20 Differentially Expressed Genes

Tree-like diagram based on how similarly genes are expressed across tumor and normal tissue samples. Genes joined by shorter branches have similar activity whereas those connected by taller branches are more different. Height represents how different the gene expression patterns are between groups

Discussion

Based on the figures and analyses, it is confirmed that molecular features of breast cancer can play a significant role in identifying cancer cells. The three figures generated to understand the key difference between the tissue samples show the mechanisms in which the tumor can grow and eventually become cancerous.

The main purpose of the PCA test was to understand and validate the difference between tumor and normal samples. From the graph, it can be understood that the clustering of the tumor and normal samples on the opposite sides of PC1 validates previous research on cancer tissues and normal tissue via PCA with respect to metabolic activity. Similarly the study illustrated normal tissues clustered on the right side and tumor tissues on the left which showed that tumor tissues and normal tissues have distinct metabolic profiles.¹ PCA is a very useful tool when understanding and visualizing differences between two groups, especially complex biological conditions.

To go more in depth, a Heatmap was generated to visualize gene activities in a tumor tissue and normal tissue as well as identify the key differences between the samples. Genes such as TSPAN6, TNMD, and CFTR have notably shown visible gene activity differences in the Heatmap. In normal tissues both TSPAN6 is upregulated compared to the tumor tissues. TSPAN6 aka Tetraspanin 6 is a protein coding gene which plays a role in cell development, activation, growth and motility.² TSPAN6 promotes secretion of oxysterol-rich extracellular vesicles which attract B cells into tumor microenvironment, however when TSPAN6 is low, B cells don't infiltrate the tumor that much which can weaken anti tumor immunity. The behavior of TSPAN6 correlated with the downregulation of the gene in tumor tissues which may have anti-tumor weakened immunity.³ TNMD and CFTR is downregulated in tumor tissues compared to the normal tissues. TNMD aka tenomodulin is responsible for stimulating chondrocyte growth and inhibiting tube formation of endothelial cells. A study has shown that lower levels of TNMD mRNA was present in tumor tissue samples compared to normal tissue samples. The study also showed that breast cancer patients with luminal B gene had higher TNMD expression in tumors hence leading to better post progression survival. CFTR aka Cystic Fibrosis Transmembrane conductance Regulator controls ion and water secretion and absorption in epithelial tissues. Mutations in CFTR have shown a higher incidence rate in breast cancer due to estrogen being a factor in affecting CFTR expression.⁴ CFTR dysfunction can also trigger impaired DNA repair which can cause mutation eventually leading to cancer.⁵

The dendrogram goes into depth about the relationships between the genes. Between the genes that showed significant gene differences mentioned previously CFTR and TNMD share a close relationship. The co-clustering of CFTR and TNMD can suggest a co-regulatory or functionally linked behavior which can represent a disruption in tumor suppressive pathways. While these genes show significant differences between normal tissue and tumor tissues, future experiments could be conducted to check for possible limitations. In vivo experiments such as developing mouse models of overexpression of TSPAN, TNMD and CFTR could be conducted to check their behavior with respect to breast cancer.

References

1. Budczies, J., Denkert, C., Müller, B. M., Brockmüller, S. F., Klauschen, F., Györfy, B., Dietel, M., Richter-Ehrenstein, C., Marten, U., Salek, R. M., Griffin, J. L., Hilvo, M., Orešič, M., Wohlgemuth, G., & Fiehn, O. (2012). Remodeling of central metabolism in invasive breast cancer compared to normal breast tissue: A GC-TOFMS based metabolomics study. *BMC Genomics*, 13(1), 334. <https://doi.org/10.1186/1471-2164-13-334>
2. National Center for Biotechnology Information. (n.d.). *TSPAN6 Homo sapiens (human) gene ID: 7105*. NCBI Gene. Retrieved June 10, 2025, from <https://www.ncbi.nlm.nih.gov/gene/7105>
3. Molostvov, G., Gachechiladze, M., Shaaban, A. M., Hayward, S., Dean, I., Dias, I. H. K., ... Berditchevski, F. (2023). Tspan6 stimulates the chemoattractive potential of breast cancer cells for B cells in an EV- and LXR-dependent manner. *Cell Reports*, 42(3), 112207. <https://doi.org/10.1016/j.celrep.2023.112207>
4. Stastna, N., Brat, K., Homola, L., Os, A., & Brancikova, D. (2023). Increasing incidence rate of breast cancer in cystic fibrosis – relationship between pathogenesis, oncogenesis and prediction of the treatment effect in the context of worse clinical outcome and prognosis of cystic fibrosis due to estrogens. *Orphanet Journal of Rare Diseases*, 18(1), 62. <https://doi.org/10.1186/s13023-023-02671-z>
5. Parisi, G. F., Papale, M., Pecora, G., Rotolo, N., Manti, S., Russo, G., & Leonardi, S. (2023). Cystic fibrosis and cancer: Unraveling the complex role of CFTR gene in cancer susceptibility. *Cancers*, 15(17), 4244. <https://doi.org/10.3390/cancers15174244>